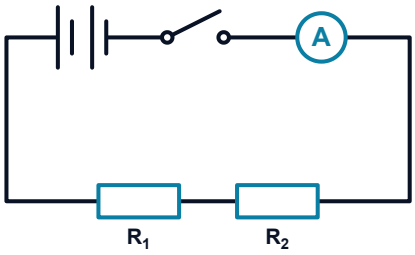
DO NOW · REQUIRED PRACTICAL

Resistors in series and in parallel

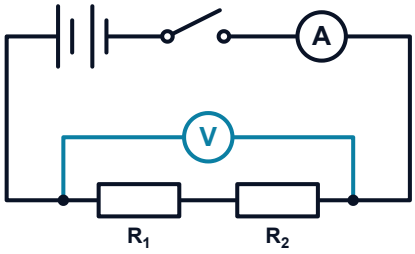
Electricity
Paper 1 · Combined and Separate
6 steps

A student compares the total resistance of two resistors in series and in parallel.

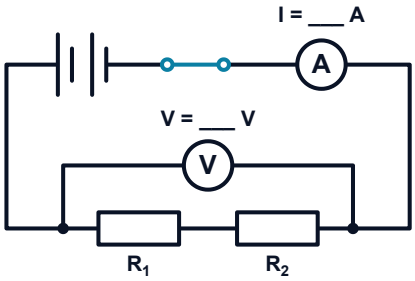
1 Connect the two resistors in series with a power supply and an ammeter.



2 Connect a voltmeter across both resistors.



3 Switch on, record the current and the potential difference, then switch off.



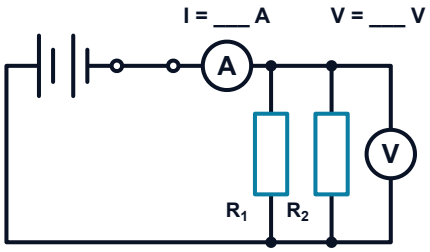
then switch off

4 Calculate the total resistance using R = V ÷ I.

R = V ÷ I
potential difference (V) ÷ current (A)

R = total resistance of the two resistors

5 Reconnect the two resistors in parallel and take the readings again.



take the readings again

6 Compare the two values of total resistance.

series	vs	parallel
R = ___ Ω		R = ___ Ω

which is bigger?

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